

Submission Summary

Conference Name

International Conference on Intelligent Computing and Sustainable Technologies

Paper ID

198

Paper Title

Privacy Preserving Edge AI Based Attention and Emotion Tracking System for Hybrid Classrooms

Abstract

The rapid growth of hybrid and online learning environments has created significant challenges in assessing student engagement and emotional responsiveness during lectures. Conventional attendance systems only measure presence and fail to capture whether students are attentive or emotionally involved in the learning process. Many existing engagement monitoring systems rely on cloud-based video analytics, which introduces concerns related to privacy, latency, and data security. This research proposes a privacy preserving Edge Artificial Intelligence system that detects student attention and emotional states in real time within hybrid classroom environments. The proposed system processes webcam video and microphone audio locally on an edge device, eliminating the need for external cloud storage and ensuring that sensitive data remains on the user device. The system integrates gaze estimation, head pose detection, facial expression recognition, and vocal tone analysis to evaluate engagement levels. Lightweight deep learning models are employed to ensure efficient computation on resource constrained devices. A multimodal fusion framework combines both visual and audio features to generate an engagement score that reflects the student's level of concentration and emotional state during the lecture. The proposed architecture provides faster response time, improved privacy protection, and scalable deployment in educational institutions. The results demonstrate that the approach can significantly enhance engagement monitoring while maintaining ethical and privacy standards.

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Data Analytics & Intelligent Applications

Secondary Subject Areas

Intelligent Systems & AI

Submission Files

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